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A Multimodal AI-Based Mobile Application for
Intelligent Household Maintenance Diagnostics and Management

Interim Presentation | June 2026

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AGENDA

1. Introduction & Background	2. Project Overview
3. System Architecture	4. Results & Progress

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01

INTRODUCTION & BACKGROUND

Understanding the household maintenance challenges in Hong Kong and the AI technologies that can address them

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THE PROBLEM: HK HOUSEHOLD MAINTENANCE

Economic Barriers

Inspection fees **HK$460-850** before repairs begin

begin

70,000 tonnes e-waste annually in HK

1,800 tonnes furniture discarded daily

Information Asymmetry

Homeowners lack knowledge to assess fault severity

severity

Cannot distinguish **minor vs major faults**

Must rely on technician judgment

Systemic Quality Issues

1,359 complaints about repair services in 2024

2024

45% involved electrical appliances

20% of consumers had contractor disputes

KEY STATISTICS DRIVING THE NEED FOR UncleFix

HK$460-850	70,000t	1,359	16%
Avg. Inspection Fee	Annual E-waste in HK	Repair Complaints (2024)	of Municipal Solid Waste

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LITERATURE: AI FOR HOME MAINTENANCE

Technology	Description	Relevance to UncleFix
Vision-Language Models	Diagnostics-LLaVA (Kumar et al., 2024) — domain-specific VLM for equipment image interpretation, fault diagnostics, and repair recommendations	Core technology for photo-based appliance identification and fault detection in the Fix screen
Multimodal Generative AI	Amazon Bedrock (Claude 3) — processes images, PDFs, videos, audio via multimodal RAG for predictive maintenance	Reference architecture for multimodal input processing and retrieval-augmented generation pipeline
RAG Frameworks	Elasticsearch-based RAG — ingests appliance manuals, performs semantic search, synthesizes context-aware answers	Directly applicable to the knowledge base retrieval and grounded answer generation
Consumer Tools	iFixit FixBot — CV-based device identification + 125,000+ repair guides; HomeCare Manager & HomeKeep apps	Validates consumer demand; no unified app exists integrating diagnostics + RAG + safety + lifecycle management

Research Gap: No unified consumer mobile application currently integrates fault diagnostics, RAG-grounded repair guidance, safety classification, and appliance lifecycle management into a single safety-conscious system tailored for household use.

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02

PROJECT OVERVIEW

Project motivation, aims, objectives, and application scope for the UncleFix mobile application

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PROJECT MOTIVATION

01 Minor Issues, Major Costs

Many common household problems — leaking taps, clogged drains, tripped breakers, appliance error codes — are minor and can often be resolved through basic troubleshooting with clear instructions.

02 Technician Dependence

Lack of technical understanding creates strong dependence on external external technicians, exposing consumers to inflated quotations, unnecessary unnecessary repairs, and limited pricing transparency.

03 Fragmented Information

Appliance manuals, warranty records, and maintenance histories are dispersed dispersed across physical and digital formats, written in technical language not language not easily understood by non-experts.

04 AI as a Solution

Recent advances in VLMs enable image-based fault identification, while RAG while RAG frameworks ensure responses are grounded in authoritative sources, authoritative sources, improving accuracy and trustworthiness.

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AIMS & OBJECTIVES

01

Integrate VLM Diagnostic Pipeline for appliance identification and visual fault/error-code extraction from user-submitted photos

02

Implement RAG Framework using a curated HK-specific knowledge base of manufacturer manuals and EMSD/GovHK regulatory guidance

03

Safety Guardrail & Severity Classification (Low / Medium / High) with hard-coded disclaimers and referral protocols for high-risk scenarios

04

iOS Asset Management Module for appliance registration, warranty tracking, and rule-based maintenance scheduling

05

Systematic Evaluation using structured benchmarks covering retrieval precision, LLM faithfulness, and VLM diagnostic accuracy across the complete UncleFix pipeline

PROJECT AIM

To design and implement a multimodal AI-powered mobile application for household diagnostics, grounded repair guidance, and appliance maintenance management — reducing management — reducing information asymmetry and promoting sustainable practices among HK homeowners.

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UncleFix: APP OVERVIEW & SCOPE

Dimension	Scope Definition
In-Scope Tasks	Minor interior household issues within safe self-assessment scope: kitchen/laundry appliance faults (washing machines, refrigerators, microwaves), minor plumbing (leaking taps, clogged drains), basic electrical resets (tripped circuit breakers)
Out-of-Scope	Structural repairs, major plumbing overhauls, gas-line configurations, building-wide infrastructural issues. For hazardous or legally regulated interventions, the app acts as a hazard-warning and referral mechanism only
System Outputs	For every diagnostic request: (1) likely fault cause and category; (2) estimated fixing time and deterministic safety risk rating (Low / Moderate / High); (3) list of tools and materials required; (4) recommended course of action with safety notes
Localization	RAG knowledge base exclusively curated from HK-specific sources: EMSD technical guidance, Consumer Council appliance data, GovHK household advisories, and manufacturer user manuals uploaded by users

Target Audience: Hong Kong residents who lack technical expertise in household damages, defects, and appliance repair, seeking reliable AI-powered guidance to diagnose issues and make informed maintenance decisions.

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03

SYSTEM ARCHITECTURE

Four-layer modular architecture: iOS frontend, FastAPI backend, AI/retrieval layer, and data storage

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ARCHITECTURE OVERVIEW

LAYER 1
iOS Presentation Layer
SwiftUI, iOS 17+

LAYER 2
API Layer
FastAPI, Python

LAYER 3
AI & Retrieval Layer
VLM + RAG + LLM

LAYER 4
Data Storage Layer
PostgreSQL + FAISS

Layer 1: Mobile Interface

- 5-screen SwiftUI app (Home, Diagnose, Appliances, History, Schedule) with dual-language support (English / Cantonese)
- Camera & photo picker integration for real-time fault diagnosis, with local notifications for maintenance reminders

Layer 2: FastAPI Backend Orchestration

- Per-request authentication (shared-secret middleware; JWT planned)
- Business-logic coordination between VLM, RAG retriever, severity classifier, and answering LLM
- Maintenance-scheduling and reminder trigger logic
- System logging, latency/cost accounting, and health probes

Layer 3: AI Components

- **VLM API:** Appliance ID, error code extraction (Moonshot Kimi K2.6)
- **RAG Pipeline:** Vector DB (FAISS), semantic retrieval of manual segments
- **Safety Guardrail:** Low / Medium / High risk classification (deterministic)
- **LLM Generation:** Structured output — fault cause + risk rating + action plan

Layer 4: Data Storage

PostgreSQL for user data, appliance registry, warranty records | FAISS for vector search | Amazon S3 for persistence

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iOS FRONTEND: FIVE CORE SCREENS

Screen	Functionality	Key Features
Home	Central navigation hub with one-tap access to all core screens	Persistent safety disclaimer: "If something feels unsafe, call a professional!"
Fix	AI-powered diagnostic interface for fault identification and repair guidance	Photo upload + issue description; structured result with severity, guidance, disclaimer
Appliances	Appliance asset management for device registration and tracking	Warranty tracking, manual upload (PDF/TEXT), color-coded warranty indicator
History	Chronological record of past maintenance activities	AI-generated repair recommendations with full diagnostic history
Reminders	Manage scheduled maintenance with push notifications	3-step task creation: select appliance → task type → recurrence and notification time

Native iOS Frameworks Used

AVFoundation	Vision	URLSession	Keychain	UserNotifications
Camera access and image capture	On-device image preprocessing	HTTPS data transmission	Secure JWT token storage	Push notification scheduling

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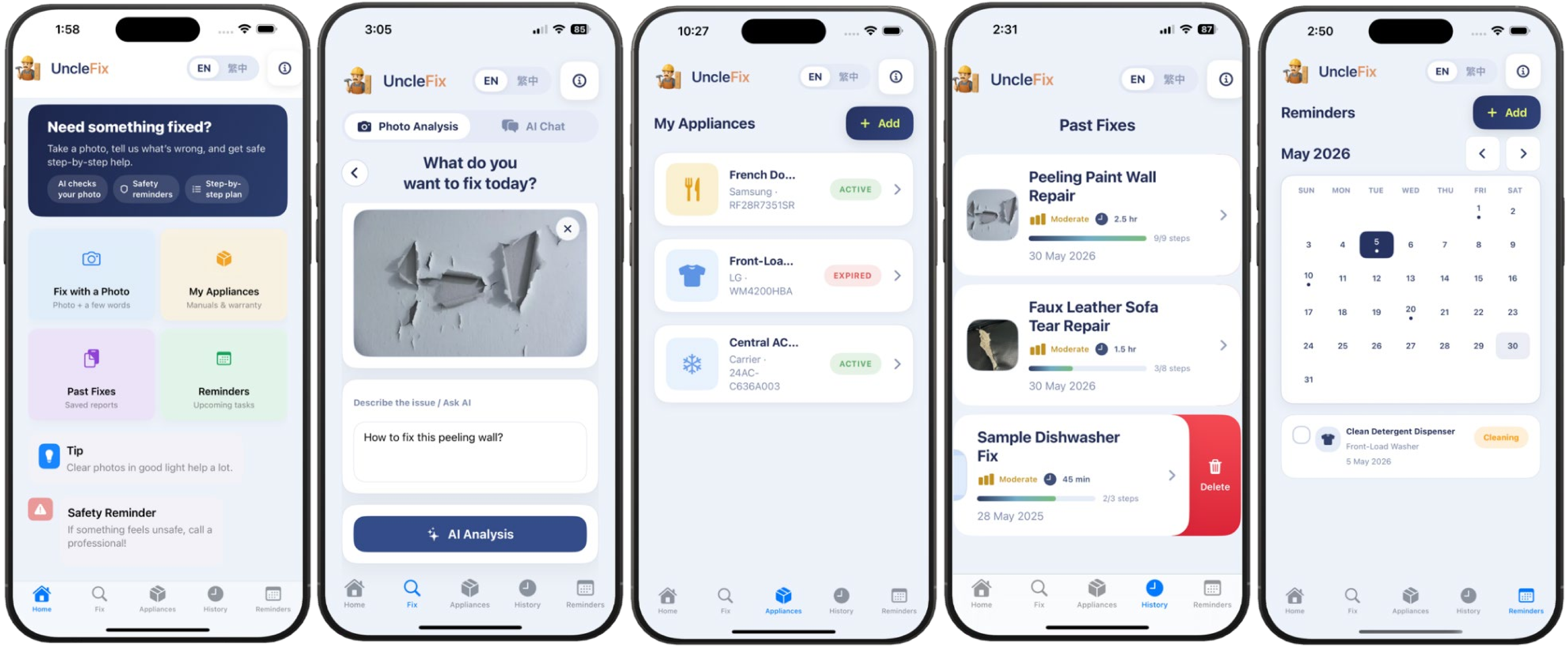

1:58
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EN 繁中
Need something fixed?
Take a photo, tell us what's wrong, and get safe step-by-step help.
AI checks your photo
Safety reminders
Step-by-step plan
Fix with a Photo
Photo + a few words
My Appliances
Manuals & warranty
Past Fixes
Saved reports
Reminders
Upcoming tasks
Tip
Clear photos in good light help a lot.
Safety Reminder
If something feels unsafe, call a professional!
Home Fix Appliances History Reminders
3:05
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EN 繁中
Photo Analysis
AI Chat
What do you want to fix today?
Describe the issue / Ask AI
How to fix this peeling wall?
AI Analysis
Home Fix Appliances History Reminders
10:27
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EN 繁中
My Appliances
+ Add
French Do...
Samsung · RF28R7351SR
ACTIVE
Front-Loa...
LG · WM4200HBA
EXPIRED
Central AC...
Carrier · 24AC-C636A003
ACTIVE
Home Fix Appliances History Reminders
2:31
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EN 繁中
Past Fixes
Peeling Paint Wall Repair
Moderate 2.5 hr
9/9 steps
30 May 2026
Faux Leather Sofa Tear Repair
Moderate 1.5 hr
3/8 steps
30 May 2026
Sample Dishwasher Fix
Moderate 45 min
2/3 steps
28 May 2025
Delete
Home Fix Appliances History Reminders
2:50
UncleFix
EN 繁中
Reminders
+ Add
May 2026
SUN MON TUE WED THU FRI SAT
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
Clean Detergent Dispenser
Front-Load Washer
5 May 2026
Cleaning
Home Fix Appliances History Reminders

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EXPERIMENT on AI PIPELINE

8 Models Tested

- Kimi-K2.6
- GLM-5.1
- GPT-5.4
- GPT-5.4-Mini
- Grok-4.2
- Qwen-3.6
- Claude-Sonnet-4.6
- Claude-Haiku-4.5

Judge on 5 Rubrics

Correctness
Completeness
Safety
Relevance
Faithfulness

Test Data

- **1,339 chunks** across **82 documents** in curated corpus
- Categorized into **Appliance Manual** and **Home Repair Guide**
- **22 Text-only test cases**
- **7 Images test cases**

Five Routing Modes — Production: Agentic (Mode C)

Mode	Name	Description
A	Pure RAG	Corpus retrieval only, no web fallback
B	Hybrid + Web Fallback	Dense + BM25 with deterministic threshold-based web search
C	Agentic (Production)	LLM decides per query which tools to invoke — corpus, web, or parametric
D	Parametric-Only	LLM's training knowledge only, no retrieval
W	Web-Only	Live web search only, no corpus retrieval

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SAFETY GUARDRAIL & KNOWLEDGE BASE

4-Signal Deterministic Safety Severity Classification

Risk Level	Examples	System Response
Low	Clogged filters, cosmetic replacements, basic resets	Step-by-step DIY guidance generated with standard safety notes
Moderate	Minor plumbing resets, external component assembly	Conditional DIY guidance with enhanced safety reminders and PPE recommendations
High	Internal electrical wiring, gas-line issues, sealed refrigerant	High-risk banner prepended; immediate professional referral protocol activated; no DIY guidance

Localized Knowledge Base Sources

Source Category	Specific Sources
HK Government	Buildings Department (drainage, water seepage, window safety), Joint FEHD & Buildings Department (DIY water seepage testing)
Consumer Council (HK)	Appliance maintenance, usage tips, safety guidance, energy-saving advice, appliance reliability data
Overseas References	Australian Government YourHome — comprehensive government-endorsed DIY maintenance guidance applicable to general household repair
Manufacturer Docs	Official manuals for washing machines, refrigerators, air conditioners, TVs, vacuum cleaners from major brands + user-uploaded manuals

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04

RESULTS & PROGRESS

Preliminary evaluation results, frontend prototype validation, and remaining project schedule

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FRONTEND PROTOTYPE: UI/UX VALIDATION

Safety-First Design

Persistent safety disclaimer on Home screen builds trust and sets user expectations before any diagnostic interaction begins

Card-Based Navigation

One-tap access to all core features, optimized for single-single-handed use with minimal navigation depth

Flow Separation

Clear separation between diagnostic flow (Fix) and management flows reduces cognitive load for non-technical technical users

Structured Results

3-component diagnostic display: severity rating + step-by-step-by-step guidance + contextual safety disclaimer

Warranty Tracking

Color-coded warranty indicator alerts users before expiry, enabling timely action on repair costs

Bilingual Support

English and Traditional Chinese across all five screens for Hong Kong's linguistically diverse population

PROTOTYPE STATUS

The SwiftUI prototype implements all five primary screens with full navigation flow, demonstrating a coherent end-to-end user journey bridging reactive diagnosis, proactive asset management, and scheduled maintenance within a unified tab-based structure. The Safety-First design philosophy is consistently expressed at the UI layer without compromising usability.

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BACKEND EVALUATION: KEY FINDINGS

880-cell evaluation — 22 golden test cases × 5 retrieval modes × 8 frontier LLMs across 6 families

FINDING 01

Agentic routing wins

LLM decides per query whether to use corpus, web, or parametric knowledge — outperforming deterministic fallback

Δ +0.366, p = 0.003

FINDING 02

Agentic uses corpus when it should

On queries with matching corpus content, agentic mode reliably recognizes and uses corpus answers

Δ +0.306, p = 0.038

FINDING 03

Parametric beats pure RAG on long-tail

When corpus is silent, frontier model training knowledge knowledge outperforms short web snippets

Δ -0.324, p = 0.006

PRODUCTION CONFIG

Agentic Mode × Kimi K2.6

Highest Safety score: **4.88 / 5**

Lowest per-query cost among top-3: **~$0.02/query**

Same model serves as both VLM and answering LLM

LATENCY GAP

Current mean latency: **30.8s** (p99: 45s)

Target: **<3s**

Mitigation: AWS Bedrock migration + streaming endpoint + answer cache

Scheduled for Phase 4 (Jun 2026)

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PROJECT STATUS & SCHEDULE

Phase	Key Tasks	Period	Status
Phase 1	Data collection, knowledge base construction, project webpage	Mar 20 – Apr 13	Completed
Progress 1	First progress update presentation	Apr 14	Completed
Phase 2	AI pipeline evaluation, multimodal RAG orchestrator, FastAPI deployment	Apr 15 – May 4	Completed
Progress 2	Second progress update presentation	May 5	Completed
Phase 3	iOS frontend development, backend integration, interim report prep	May 6 – May 31	Completed
Interim	Interim report submission and presentation	Jun 1, 2026	In Progress
Phase 4	AWS Bedrock migration, PostgreSQL integration, safety guardrail refinement	Jun 2 – Jun 16	Pending
Phase 5	Functional testing (60+ scenarios), usability testing (10-15 users)	Jun 17 – Jul 6	Pending
Phase 6	Documentation, demo video, project webpage, final dissertation	Jul 7 – Jul 17	Pending
Final	Final report submission and oral examination	Jul 17 – Late Jul	Pending

THANK YOU!

UncleFix — Empowering Hong Kong Homeowners

with AI-Driven Maintenance Intelligence

Questions & Discussion